09-22 第1篇

阅读理解（说明文） Why Does the Moon Change Shape Every Night?

① On the night of the **Mid-Autumn Festival**, families across China look up and see a large, perfectly round **full moon**. Two weeks later, the same Moon looks like a thin silver "**C**". Two weeks after that, it almost disappears from the sky. Has the Moon suddenly become smaller, or broken into pieces? Of course not. The Moon is always the same size; it is **our view of it** that keeps changing. So why, then, does the Moon appear in so many different shapes across the month?

② The first important fact is that the Moon **does not make its own light**. It only **reflects sunlight**, like a huge grey mirror floating in space. This means that at any moment, **one half** of the Moon is always bright (facing the Sun), and **the other half** is always dark (turned away from the Sun). The Moon's shape, from our view, depends entirely on **how much of the bright half we can see from Earth**.

③ The second important fact is that the Moon **travels around the Earth**, completing one circle about every **29.5 days**. As the Moon moves, the angle between the Sun, the Moon and the Earth keeps changing. When the Moon is on the **same side** of the Earth as the Sun, the bright half faces the Sun and the dark half faces us — we see almost nothing, and we call this a **"new moon"**. When the Moon is on the **opposite side**, the Sun lights up the whole side facing us, and we see the full bright disc — a **"full moon"**.

④ Between these two extremes, the Moon passes through several familiar stages, called **phases**. A few days after the new moon, a thin bright line appears on the right edge of the Moon — the **crescent moon**, shaped like a narrow "C" (in the Northern Hemisphere the bright side is usually on the right). About a week later, we see **half** of the bright side — this is the **first quarter** moon, which looks like the letter **D**. A few days more, and most of the bright side is visible — the **gibbous moon**, nearly full. After the full moon, the whole process **reverses**: gibbous, then third quarter (like a backwards D), then waning crescent, then new moon again. The whole cycle takes **about one month** — in fact, the word "month" comes from "Moon".

⑤ Many people confuse the **phases of the Moon** with **eclipses**. They are not the same. **Phases** happen every month, quietly, because of the Moon's normal journey. An **eclipse** happens only a few times a year, and only when the Sun, Earth and Moon line up in an almost perfect straight line. During a **lunar eclipse**, the Earth's shadow falls across the Moon and turns it reddish for a short time; during a **solar eclipse**, the Moon passes in front of the Sun and casts a shadow on the Earth. Eclipses are **rare and dramatic**; phases are regular and gentle.

⑥ A practical tip for young moon watchers: if you see a **"D-shaped" moon**, the Moon is **growing** and will be full in about a week; if you see a **backwards-D shape** (a "C" that opens to the right), the Moon is **shrinking** and will soon disappear. In English there are two neat words for this: a **waxing** moon is getting bigger, a **waning** moon is getting smaller. Once you learn these two words, even a single quick look at the sky will tell you which part of the month it is.

⑦ So, the next time a small child asks "Where did the rest of the Moon go?" you can smile and say: nowhere. The Moon is still whole, still about **3 474 km** across, still **384 000 km** from us, still turning quietly around the Earth as it has for more than four billion years. What changes each night is only the **light** and our **angle of view** — a small reminder that in astronomy, as in life, what we see and what is truly there are not always the same thing.

注释： reflect 反射；phase 月相；crescent 月牙；gibbous 凸月；eclipse 食（日/月食）；waxing 盈；waning 亏

根据文章内容，选择最佳答案。

1. What is the main question the passage tries to answer?

A. How big the Moon actually is.
B. Why the Moon appears in different shapes across the month.
C. Whether people can live on the Moon.
D. What makes the Sun bright.

2. What is the first important fact about the Moon in Paragraph 2?

A. The Moon makes its own light.
B. The Moon does not make its own light; it only reflects sunlight, so one half is always bright and the other is always dark.
C. The Moon is always fully lit by the Sun.
D. The Moon shines most brightly in the morning.

3. About how long does the Moon take to complete one cycle of phases?

A. About 24 hours.
B. About 7 days.
C. About 29.5 days, roughly one month.
D. About one year.

4. Which statement correctly distinguishes "phases" from "eclipses" (Paragraph 5)?

A. Phases and eclipses are the same thing.
B. Phases happen every month because of the Moon's normal journey; eclipses happen

only a few times a year and need an almost perfect straight line of Sun—Earth—Moon.
C. Eclipses happen every night, but phases are rare.
D. Only eclipses make the Moon look round.

5. If you see a bright "D-shaped" Moon in the sky, what does it mean?

A. The Moon is shrinking and will soon disappear.
B. The Moon is waxing — growing — and will be full in about a week.
C. The Moon has broken into pieces.
D. There will be a solar eclipse tomorrow.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】第①段 "why does the Moon appear in so many different shapes across the month?"
【解析】全文解释月相变化。A、C、D 与主题不符。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "does not make its own light ... reflects sunlight ... one half is always bright"
【解析】月不自发光，仅反射阳光。A、C、D 均错。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第③段 "completing one circle about every 29.5 days"
【解析】一个月相周期约 29.5 天。A、B、D 均错。故选 C。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段 "Phases happen every month ... eclipse ... only when ... almost perfect straight line"
【解析】月相常见有规律，食则罕见需三点共线。A、C、D 均错。故选 B。

5.
【答案】B
【题型】细节理解题

【定位】第⑥段 "D-shaped moon ... growing ... will be full in about a week ... waxing"
【解析】D 形月处于上弦、走向满月。A、C、D 均错。故选 B。